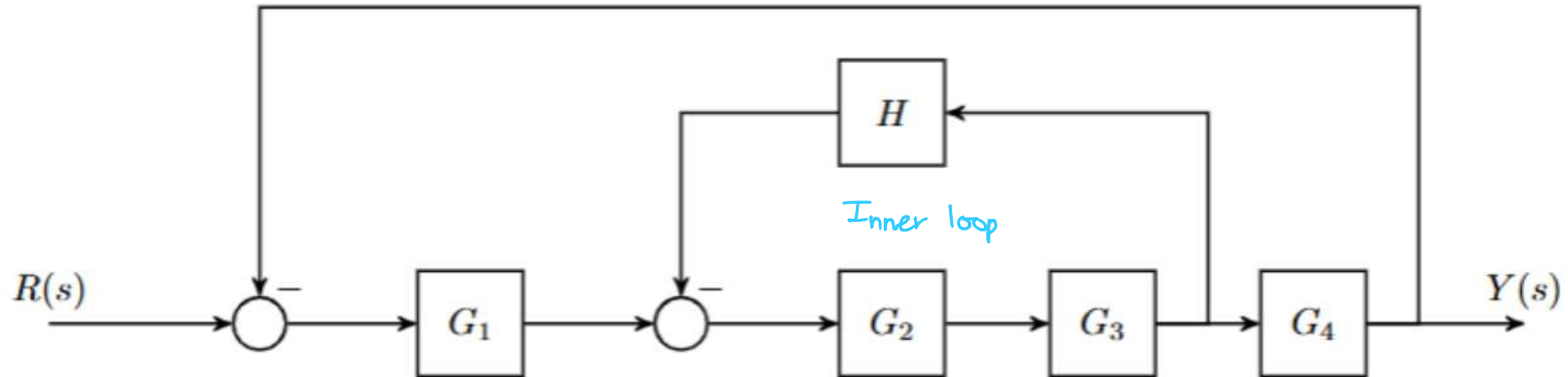


Homework 2

Wednesday, September 16, 2026 11:09 AM

1. Given:



Find:

- The general open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- Set $G_1 = \frac{1}{s}$, $G_2 = \frac{1}{s+1}$, $G_3 = \frac{1}{s}$, $G_4 = \frac{2}{s+2}$, and $H = \frac{5}{s+5}$ then determine the open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

a) For the inner loop,

$$G_{inner} = \frac{G_2 G_3}{1 + G_2 G_3 H}$$

The forward path seen by the outer loop:

$$G_F = G_1 \left(\frac{G_2 G_3}{1 + G_2 G_3 H} \right) G_4$$

With the outer unity negative feedback,

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$$\frac{Y(s)}{R(s)} = \frac{G_f}{1+G_f} = \frac{G_1 G_2 G_3 G_4}{1+G_2 G_3 G_4 + G_1 G_2 G_3 G_4} = \frac{B(s)}{A(s)}$$

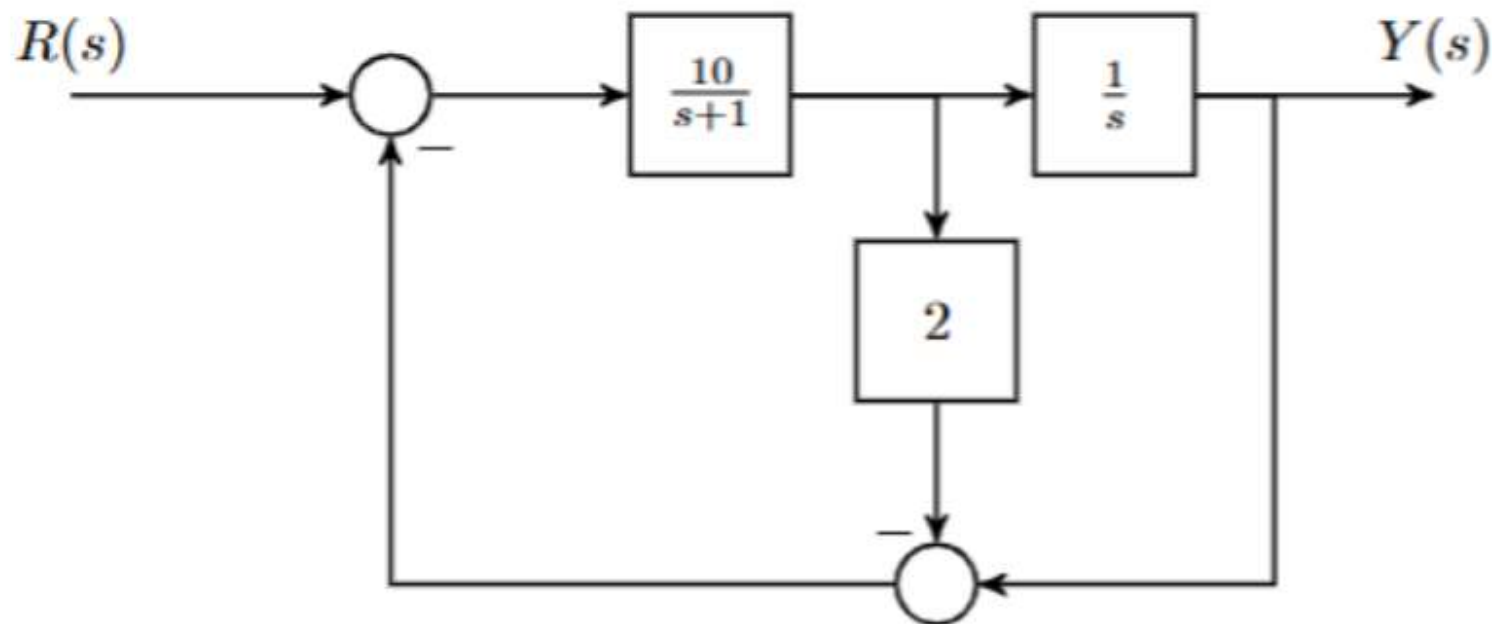
$$b) G_1 G_2 G_3 G_4 = \frac{5}{s^2(s+1)(s+2)}$$

$$G_2 G_3 H = \frac{5}{s(s+1)(s+5)}$$

$$\frac{Y}{R} = \frac{\frac{5}{s^2(s+1)(s+2)}}{1 + \frac{5}{s(s+1)(s+5)} + \frac{5}{s^2(s+1)(s+2)}} = \frac{5(s+5)}{s^5 + 8s^4 + 17s^3 + 15s^2 + 15s + 25}$$

$$B(s) = 5s + 25; \quad A(s) = s^5 + 8s^4 + 17s^3 + 15s^2 + 15s + 25$$

2. Given:



Find:

(a) Open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

(b) Closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

Let $X(s)$ be the signal immediately after $\frac{10}{s+1}$.

Since the integrator follows it, $Y(s) = \frac{1}{s}X(s)$, so $X(s) = sY(s)$.

At the lower summing junction, the feedback signal is $Y-2X$.

Therefore, $H(s)Y(s) = Y(s) - 2sY(s)$ and hence, $H(s) = 1-2s$.

The forward path is $G(s) = \frac{10}{s+1}$.

Therefore, $H(s)Y(s) = Y(s) - 2sY(s)$ and hence, $H(s) = 1 - 2s$.

The forward path is $G(s) = \left(\frac{10}{s+1}\right) \left(\frac{1}{s}\right) = \frac{10}{s(s+1)}$.

a) Using $\frac{Y}{R} = \frac{G}{1 + GH}$ gives $\frac{Y}{R} = \frac{\frac{10}{s(s+1)}}{1 + \frac{10}{s(s+1)}(1-2s)}$.

Multiply through by $s(s+1)$: $\frac{Y}{R} = \frac{10}{s(s+1) + 10(1-2s)} = \frac{10}{s^2 + s + 10 - 20s} \Rightarrow \frac{10}{s^2 - 19s + 10}$

$B(s) = 10$

$A(s) = s^2 - 19s + 10$

b) $G(s) = \frac{10}{s+1}$; $H(s) = 1 - 2s$

$\frac{G}{1 + GH} = \frac{10}{s^2 - 19s + 10}$

3. Given:

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2 + 30s + 200}$$

(a) Find $y(\infty)$ when the input is:

- i. $u(t)$ is an impulse
- ii. $u(t)$ is a unit step

(b) Determine a block diagram of the equation

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2+30s+200} \Rightarrow \frac{4(s+50)}{(s+10)(s+20)}$$

Both poles are in the left half-plane, so the system is stable and Final Value Theorem can be used.

(a-i) For an impulse, $u(t) = \delta(t)$ and $U(s) = 1$.

$$\text{Thus, } Y(s) = \frac{4(s+50)}{(s+10)(s+20)}$$

$$\text{Using the Final Value Theorem, } y(\infty) = \lim_{s \rightarrow 0} sY(s) \Rightarrow \lim_{s \rightarrow 0} \frac{4s(s+50)}{(s+10)(s+20)}$$

$$\text{Therefore, } y(\infty) = 0.$$

(a-ii) For a unit step, $U(s) = \frac{1}{s}$.

$$\text{Therefore, } Y(s) = \frac{4(s+50)}{s(s^2+30s+200)}$$

Final Value Theorem:

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) \Rightarrow \lim_{s \rightarrow 0} \frac{4(s+50)}{s^2+30s+200} \Rightarrow \frac{4(50)}{200} = 1$$

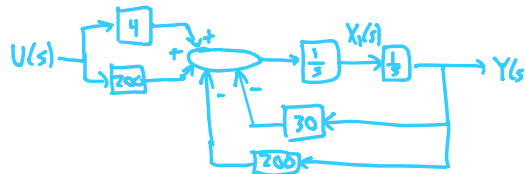
$$y(\infty) = 1$$

$$\text{b) Starting with } \frac{Y(s)}{U(s)} = \frac{4s+200}{s^2+30s+200}$$

$$(s^2+30s+200)Y(s) = (4s+200)U(s)$$

$$s^2Y(s) + 30sY + 200Y = 4sU + 200U$$

$$\ddot{y} + 30\dot{y} + 200y = 4\dot{u} + 200u$$



4. Given:

$$\frac{Y(s)}{U(s)} = \frac{s^2 - 4s + 6}{s^2 + 3s + 6}$$

(a) Find $y(\infty)$ when the input is:

i. $u(t)$ is an impulse

ii. $u(t)$ is a unit step

(b) Find a block diagram of the equation

$$\frac{Y(s)}{U(s)} = \frac{s^2 - 4s + 6}{s^2 + 3s + 6}$$

$$s = \frac{-3 \pm \sqrt{9 - 24}}{2} = \frac{-3 \pm j\sqrt{15}}{2}$$

Both have negative real parts, so the system is stable.

$$(a-i) \quad U(s) = 1$$

The transient portion of the impulse response decays to 0.

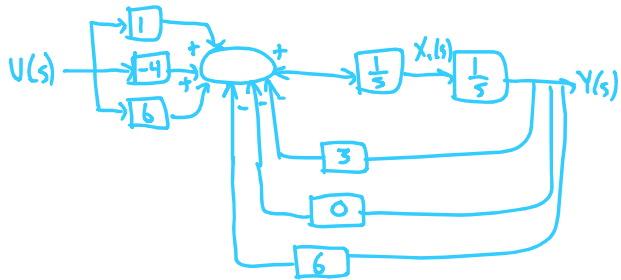
$$y(\infty) = 0$$

$$(a-ii) \quad U(s) = \frac{1}{s}$$

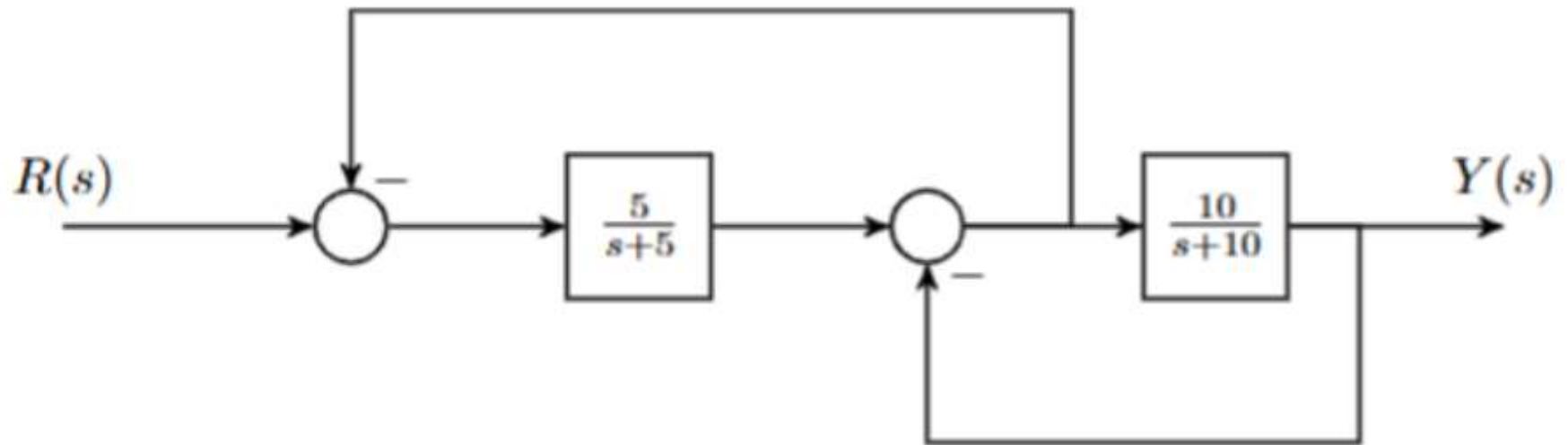
$$\text{Using the Final Value Theorem, } y(\infty) = \lim_{s \rightarrow 0} s \left[\frac{s^2 - 4s + 6}{s(s^2 + 3s + 6)} \right] = \frac{6}{6} = 1$$

Using the Final Value Theorem, $y(\infty) = \lim_{s \rightarrow 0} s \left[\frac{s^2 - 4s + 6}{s(s^2 + 3s + 6)} \right] = \frac{6}{6} = 1$
 $y(\infty) = 1$.

b) $(s^2 + 3s + 6)Y = (s^2 - 4s + 6)U$
 $s^2 Y + 3sY + 6Y = s^2 U - 4sU + 6U$
 $\ddot{y} + 3\dot{y} + 6y = \ddot{u} - 4\dot{u} + 6u$



5. Given:



Find:

(a) The transfer function $\frac{Y(s)}{R(s)}$

(b) Closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

Start w/ the $\frac{10}{s+10}$ inner loop.

It has unity negative feedback, so:

$$G_{\text{inner}} = \frac{\frac{10}{s+10}}{1 + \frac{10}{s+10}} = \frac{10}{s+20}$$

The forward path from the first summing junction to Y is consequently $G(s) = \frac{5}{s+5} \left(\frac{10}{s+20} \right) = \frac{50}{(s+5)(s+20)}$

The outer feedback signal is taken before the $\frac{10}{s+10}$ block.

Since $Y = \frac{10}{s+10} X$, then $X = \frac{s+10}{10} Y$. Therefore $H(s) = \frac{s+10}{10}$

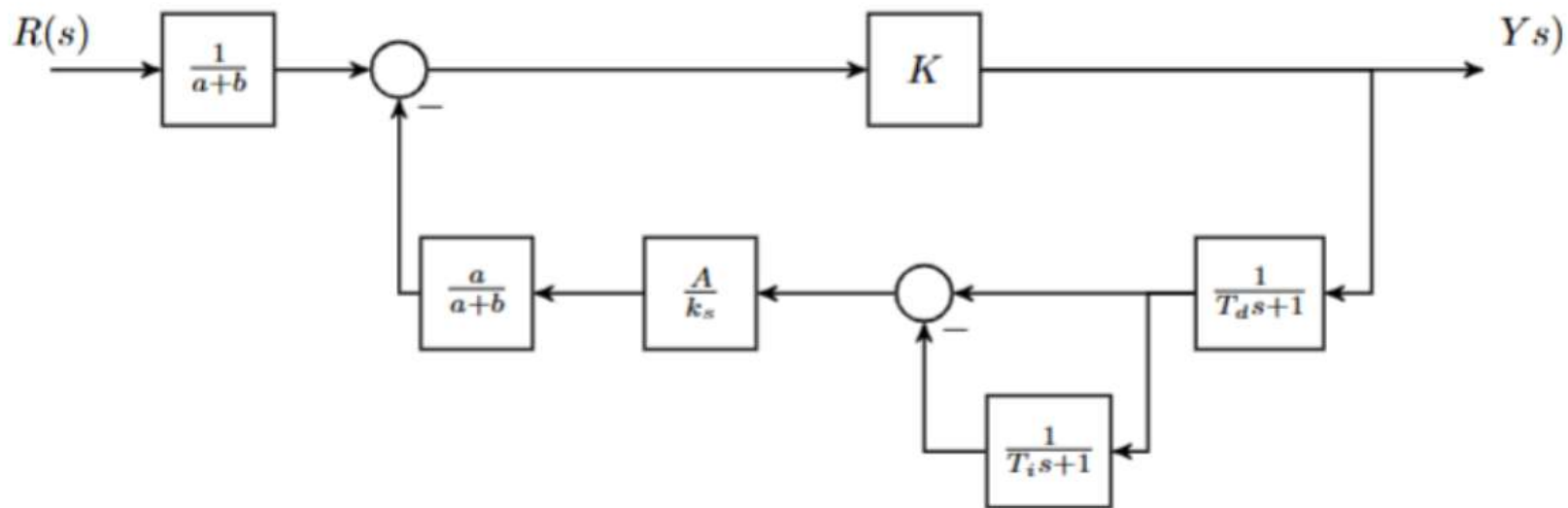
the outer feedback signal is taken before the $\frac{10}{s+10}$ block.

Since $Y = \frac{10}{s+10} X$, then $X = \frac{s+10}{10} Y$. Therefore, $H(s) = \frac{s+10}{10}$.

$$a) \frac{Y}{R} = \frac{G}{1+GH} \Rightarrow \frac{\frac{50}{(s+5)(s+20)}}{1 + \frac{50}{(s+5)(s+20)} \left(\frac{s+10}{10} \right)} \Rightarrow \frac{50}{(s+5)(s+20) + 5(s+10)} \Rightarrow \frac{50}{s^2 + 30s + 150}$$

$$b) G(s) = \frac{50}{(s+5)(s+20)}, H(s) = \frac{s+10}{10}$$

6. Given:



Find:

(a) The transfer function, $\frac{Y(s)}{R(s)}$

(b) Set: $a = 1$, $b = 3$, $T_i = 10$, $T_d = 1$, $A = 2$, $k_s = 4$, and $K = 1$ then find the poles and zeros of the system.

a) The input first passes through $\frac{1}{a+b}$.

The forward gain after the main summing junction is K .

Consider the feedback path.

the forward gain and the lower summing junction.

Consider the feedback path.

From $Y(s)$, the first block is $\frac{1}{T_d s + 1}$

Call the output $X(s)$.

At the lower summing junction, X is combined with the negative signal $\frac{1}{T_d s + 1} X$.

Thus its output is $X - \frac{X}{T_d s + 1} \Rightarrow X \left(1 - \frac{1}{T_d s + 1}\right) \Rightarrow X \frac{T_d s}{T_d s + 1}$.

Therefore, the total feedback transfer function is $H(s) = \frac{1}{T_d s + 1} \left(\frac{T_d s}{T_d s + 1}\right) \left(\frac{A}{K_s}\right) \left(\frac{a}{a+b}\right)$

$$H(s) = \frac{A a T_d s}{K_s (a+b) (T_d s + 1) (T_i s + 1)}$$

The signal entering the main summer from the reference side is $\frac{R(s)}{a+b}$.

Thus, $Y = K \left[\frac{R}{a+b} - H Y \right]$.

$$\frac{Y(s)}{R(s)} = \frac{\frac{K}{a+b}}{1 + \frac{K A a T_d s}{K_s (a+b) (T_d s + 1) (T_i s + 1)}}$$

$$b) \frac{Y}{R} = \frac{4(s+1)(s+10)}{4(4)(s+1)(10s+1) + 1(2)(1)(10)s} \Rightarrow \frac{4(s+1)(10s+1)}{4(s+1)(10s+1) + 5s}$$

$$\frac{Y(s)}{R(s)} = \frac{10s^2 + 11s + 1}{40s^2 + 49s + 4}$$

Zeros: $10s^2 + 11s + 1 = 0$

$$z_1 = -1, z_2 = -0.1$$

Poles: $40s^2 + 49s + 4 = 0$

$$p_1 \approx -1.137053$$

$$P_1 \approx -0.087947$$